

Facility / evidence

M100 closed / frozen



Frontier QC closeout



Lei-Masanet annual/RNG gate



Modern AI IT-power experiment

Water / groundwater

Identify source-water bridge Ψ



Pumping $\rightarrow$ GW chronological benchmark



Reduced-order GW module
only if benchmark supports it

Integrated planning

M_0 static PSCC-style baseline



M_1 archetypes + source/GW
only if prior gates pass



M_2 higher fidelity
only if decision-relevant

Fail a gate: restrict claims; do not escalate or retune through the failure.

Final escalation, only after coupling is credible: decision replay / regret $\rightarrow$ uncertainty $\rightarrow$ policy / decentralization